CardioVision AI

Multi-Vessel Risk Assessment & Clinical Vector Evaluation

RESEARCH PROTOTYPE EVALUATION REPORT — NOT FOR CLINICAL DIAGNOSIS

Evaluation Date: Tue, Oct 13, 2026, 09:17 PM
Patient ID: CV-HEALTHY-01
Architecture: 57-Feature Four-Model Architecture
Active Mode: Standard Production Mode

PATIENT NAME	PATIENT ID	AGE / BIOLOGICAL SEX	HEIGHT / WEIGHT / BMI
Jordan Taylor (Low-Risk Profile)	**CV-HEALTHY-01**	**28 yrs · Female**	**165 cm / 55 kg (20.2 kg/m²)**

1. Cardiovascular Prediction Summary

Multi-vessel predictive classification derived from the 57-dimensional clinical feature vector.

OVERALL CAD	LAD ARTERY	LCX ARTERY	RCA ARTERY
1.7%	**52.0%**	**16.0%**	**2.8%**
Normal	**Stenotic**	**Normal**	**Normal**
cad_model.pkl Threshold: **50.0%**	lad_model.pkl Threshold: **50.0%**	lcx_model.pkl Threshold: **50.0%**	rca_model.pkl Threshold: **50.0%**
Low Est. Prob (<50%)	**Moderate Est. Prob (50–69%)**	**Low Est. Prob (<50%)**	**Low Est. Prob (<50%)**

Model Estimation Notice: Values are model-estimated statistical classification probabilities, not clinically calibrated epidemiological risk scores or measured anatomical luminal stenosis narrowing percentages.

2. Clinical Input Measurements Summary

Patient clinical parameters from active evaluation session evaluated against informational reference benchmarks.

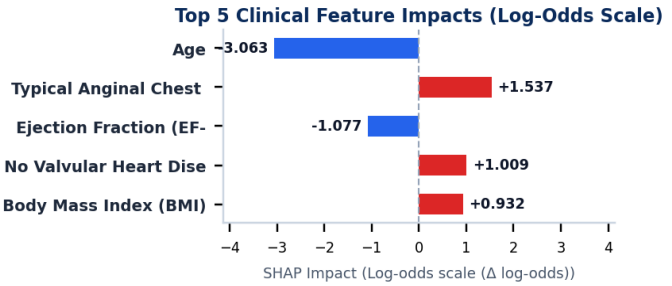
Clinical Parameter	Patient Value	General Reference Benchmark	Informational Evaluation Status
Resting Systolic BP (BP)	**110 mmHg**	< 120 mmHg (resting systolic)	Within reference (< 120 mmHg)
Resting Pulse Rate (PR)	**65 bpm**	60 – 100 bpm	Within standard resting range
Left Ventricular EF (EF-TTE)	**62%**	≥ 50%	LVEF ≥ 50% (Standard systolic function)
Fasting Blood Sugar (FBS)	**82 mg/dL**	70 – 99 mg/dL	Within fasting range (70–99 mg/dL)
Serum Creatinine (CR)	**0.8 mg/dL**	0.7 – 1.3 mg/dL	Within displayed adult reference range (Informational)
LDL Cholesterol	**75 mg/dL**	< 100 mg/dL	Within optimal reference (< 100 mg/dL)
HDL Cholesterol	**65 mg/dL**	> 40 mg/dL (M) / > 50 mg/dL (F)	Within reference (> 40 mg/dL)
Triglycerides (TG)	**85 mg/dL**	< 150 mg/dL	Within reference (< 150 mg/dL)
Evaluated Clinical Vector	**57 Model Inputs**	Standardized 7-domain vector	34 numeric + 23 binary clinical indicators synchronized

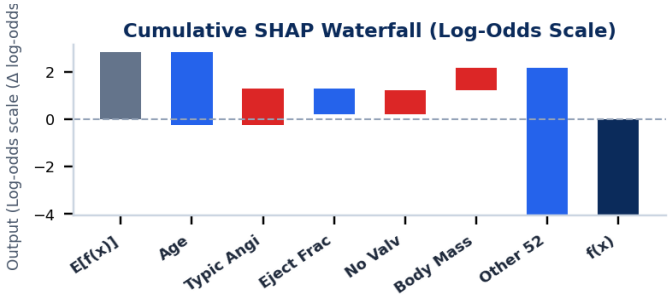
3. Patient-Specific SHAP Explainability (CAD & LAD)

Feature contributions calculated via game-theoretic Shapley values. **Scale Note:** TreeExplainer outputs additive probability contributions (Δ probability). LinearExplainer outputs additive log-odds contributions (Δ log-odds). Scales are strictly separated.

CAD Model Explainability (cad_model.pkl) · Method: **LinearExplainer (Linear SHAP)** · Target: **CAD** · **[LOG-ODDS SCALE]**
Expected Baseline E[f(x)]: **2.810 log-odds** | Final Patient Output f(x): **-4.055 log-odds** | **✓ Additivity Verified: E[f(x)] + Σ ϕ_i = f(x)**

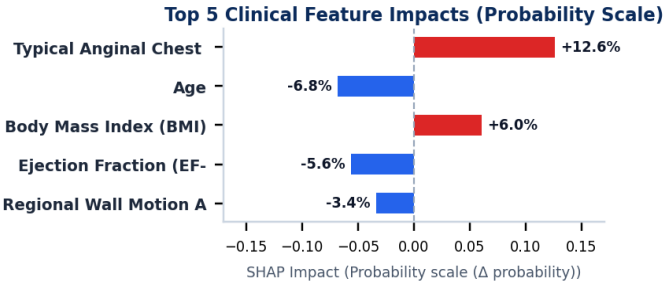
Rank	Clinical Feature	Category	Patient Value	SHAP (log-odds)	Impact Direction
#1	Age	Basic Demographics	28 years	-3.063 log-odds	Decreases Risk
#2	Typical Anginal Chest Pain	Symptoms	Present (1)	+1.537 log-odds	Increases Risk
#3	Ejection Fraction (EF-TTE)	Vital Signs	62 %	-1.077 log-odds	Decreases Risk
#4	No Valvular Heart Disease	Valvular Disease	Present (1)	+1.009 log-odds	Increases Risk
#5	Body Mass Index (BMI)	Basic Demographics	20.2 kg/m²	+0.932 log-odds	Increases Risk

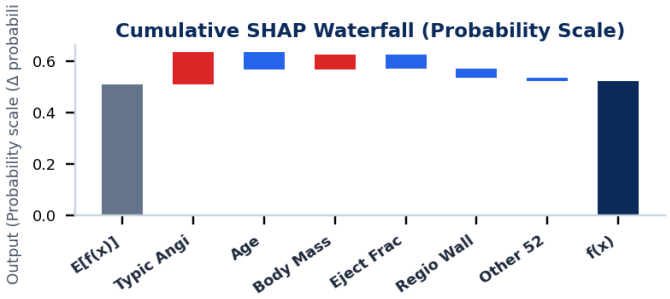




LAD Artery Model Explainability (lad_model.pkl) · Method: **TreeExplainer (Tree SHAP)** · Target: **Stenotic** · **[PROBABILITY SCALE]**
Expected Baseline E[f(x)]: **50.7%** | Final Patient Output f(x): **52.0%** | **✓ Additivity Verified: E[f(x)] + Σ ϕ_i = f(x)**

Rank	Clinical Feature	Category	Patient Value	SHAP (probability)	Impact Direction
#1	Typical Anginal Chest Pain	Symptoms	Present (1)	+12.6%	Increases Risk
#2	Age	Basic Demographics	28 years	-6.8%	Decreases Risk
#3	Body Mass Index (BMI)	Basic Demographics	20.2 kg/m²	+6.0%	Increases Risk
#4	Ejection Fraction (EF-TTE)	Vital Signs	62 %	-5.6%	Decreases Risk
#5	Regional Wall Motion Abnormality	Echocardiography	Absent (0)	-3.4%	Decreases Risk



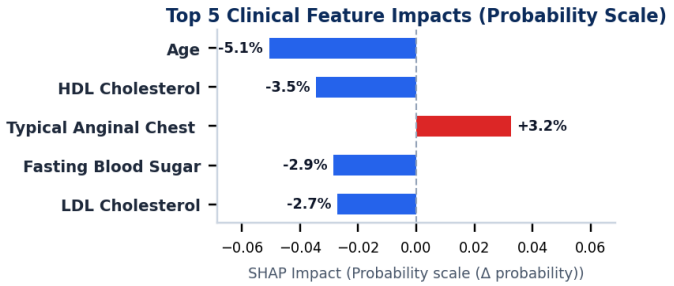


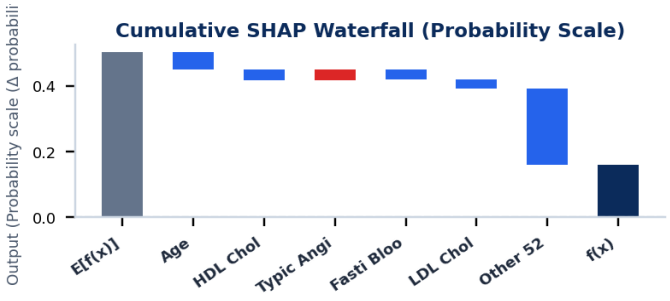
3. Patient-Specific SHAP Explainability (LCX & RCA)

Candidate and baseline branch vessel models with explicit scale separation and additivity verification.

LCX Artery Model Explainability (lcx_model.pkl) · Method: **TreeExplainer (Tree SHAP)** · Target: **Stenotic** · **[PROBABILITY SCALE]**
Expected Baseline E[f(x)]: **50.2%** | Final Patient Output f(x): **16.0%** | **✓ Additivity Verified: E[f(x)] + Σ ϕ_i = f(x)**

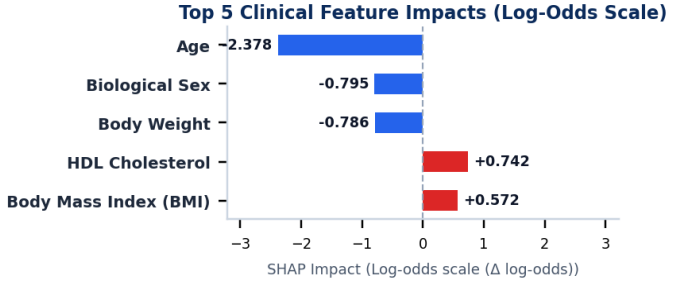
Rank	Clinical Feature	Category	Patient Value	SHAP (probability)	Impact Direction
#1	Age	Basic Demographics	28 years	-5.1%	Decreases Risk
#2	HDL Cholesterol	Biomarkers	65 mg/dL	-3.5%	Decreases Risk
#3	Typical Anginal Chest Pain	Symptoms	Present (1)	+3.2%	Increases Risk
#4	Fasting Blood Sugar	Biomarkers	82 mg/dL	-2.9%	Decreases Risk
#5	LDL Cholesterol	Biomarkers	75 mg/dL	-2.7%	Decreases Risk

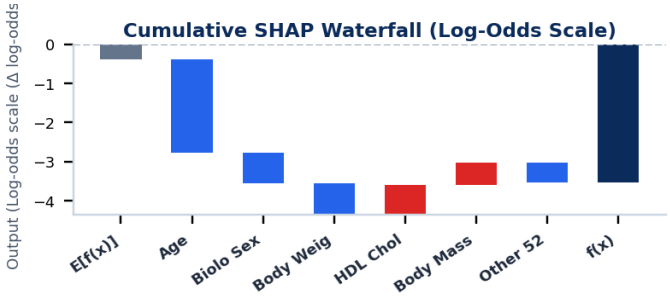




RCA Artery Model Explainability (rca_model.pkl) · Method: **LinearExplainer (Linear SHAP)** · Target: **Stenotic** · **[LOG-ODDS SCALE]**
Expected Baseline E[f(x)]: **-0.393 log-odds** | Final Patient Output f(x): **-3.532 log-odds** | **✓ Additivity Verified: E[f(x)] + Σ ϕ_i = f(x)**

Rank	Clinical Feature	Category	Patient Value	SHAP (log-odds)	Impact Direction
#1	Age	Basic Demographics	28 years	-2.378 log-odds	Decreases Risk
#2	Biological Sex	Basic Demographics	Female	-0.795 log-odds	Decreases Risk
#3	Body Weight	Basic Demographics	55 kg	-0.786 log-odds	Decreases Risk
#4	HDL Cholesterol	Biomarkers	65 mg/dL	+0.742 log-odds	Increases Risk
#5	Body Mass Index (BMI)	Basic Demographics	20.2 kg/m²	+0.572 log-odds	Increases Risk





4. 3D Coronary Heart Visualization

High-resolution anatomical perspective from the interactive Three.js 106-mesh BodyParts3D cardiac model.



5. Model Performance and Reliability

Standard Mode: Retrospective model evaluation metrics recorded during development on held-out test data.

Model	Active Model File	Target Vascular Outcome	Accuracy	F1 Score	ROC-AUC
CAD Model	cad_model.pkl	Overall Coronary Artery Disease	**85.2%**	89.7%	**0.917**
LAD Model	lad_model.pkl	Left Anterior Descending Stenosis	**65.6%**	67.7%	**0.756**
LCX Model	lcx_model.pkl	Left Circumflex Stenosis	**57.4%**	53.6%	0.584
RCA Model	rca_model.pkl	Right Coronary Artery Stenosis	**60.7%**	50.0%	0.652

Model Reliability Notice: Standard LCX and RCA baseline models exhibit limited discriminatory power in development cohort testing (ROC-AUC 0.584 and 0.652). Probabilities must be interpreted with extreme caution and never guide clinical interventions independently.

6. Research Limitations, Educational Guidance & Disclaimer

Scope of Research Evaluation & Clinical Considerations:
- Model estimates represent statistical pattern matching against retrospective cohorts, not definitive clinical diagnoses.
- Results must always be interpreted in full clinical context by a licensed medical practitioner.
- Machine learning probabilities cannot substitute for direct coronary angiography, CT angiography, or stress imaging.
- Diagnostic and management decisions depend on patient presentation, physical exam, and comprehensive clinical history.

RESEARCH PROTOTYPE — NOT A CERTIFIED MEDICAL DEVICE:
This software and report are intended strictly for educational, scientific research, and methodology evaluation purposes. They are not approved or cleared by any regulatory health agency (e.g., FDA, CE, PMDA) for clinical diagnostic or therapeutic decision-making. Predictions and SHAP explainability analyses do not substitute for formal cardiovascular imaging, laboratory testing, or physician consultation.